



UNIVERSITY OF SCIENCE - VNUHCM

FACULTY OF INFORMATION TECHNOLOGY

SOFTWARE TESTING

HW3 - DATA GENERATION

Software Testing Project Report

Authors:

Trần Thị Cát Tường
(22127444)

ttctuong22@clc.fitus.edu.vn

Supervisors:

Teacher Trần Duy Hoàng

Teacher Hồ Tuấn Thanh

Teacher Trương Phước Lộc

June 22, 2025

Table of Contents

1	Group Information	1
2	Data Generation	2
2.1	Data tables and ranges of rules	2
2.1.1	Table Users	2
2.1.2	Table Contact Requests	3
2.2	Data Generation Tools and Process	3
2.2.1	Tools Used	3
2.2.2	Generation Process	4
2.2.3	Sample Data	7
3	Self-Evaluation (Data Generation)	8

1 Group Information

Group ID: 07

Member Name	Student ID	Assigned Tables	Status
Cao Uyển Nhi	22127310	- Invoices	Done
		- Invoice Items	Done
Lưu Thanh Thuý	22127410	- Brands	Done
		- Categories	Done
Nguyễn Phước Minh Trí	22127424	- Product Image	Done
		- Personal Access Token	Done
Võ Lê Việt Tú	22127435	- Product	Done
		- Favorites	Done
Trần Thị Cát Tường	22127444	- User	Done
		- Contact Requests	Done

2 Data Generation

2.1 Data tables and ranges of rules

To ensure that the generated data is valid and usable in the system, all field rules were derived from three main sources:

- **Laravel migration files** were used to determine the technical constraints of each field, such as data type, maximum length, nullability, and uniqueness.
- **The user interface (UI)** provided insights into how fields are used in practice. Some constraints, like the **subject** field in the **contact_requests** table appear as a dropdown in the UI. This indicates a predefined set of values (an implicit enum) that should be respected during data generation.
- **Business rules and domain knowledge** were applied to define logical ranges and relationships between fields. Similarly, **updated_at** was always generated to be equal to or later than **created_at** to maintain timestamp consistency.

2.1.1 Table Users

Field Name	Type	Rule for Valid Data Generation
first_name	string(40)	Common given names (≤ 40 characters)
last_name	string(20)	Common surnames (≤ 20 characters)
address	string(70)	Short full addresses (≤ 70 characters)
city	string(40)	Real cities (≤ 40 characters)
state	string(40)	Optional; can be blank or a valid state name
country	string(40)	One of: "Vietnam", "USA", "UK", etc.
postcode	string(10)	Optional; 5–6 digit ZIP/postal code or blank
phone	string(24)	Optional; numeric string 9–11 digits, or blank
dob	date	Between 1960–2010 (age 15–65)
email	string(60), unique	Unique email, valid format, ≤ 60 characters
password	string	Optional; alphanumeric or hashed string
role	string	One of: "customer", "admin"
enabled	boolean, default true	Either true or false
failed_login_attempts	integer, default 0	Integer ≥ 0
created_at, updated_at	timestamps	ISO timestamp format, updated_at $\geq$ created_at

2.1.2 Table Contact Requests

Field Name	Type	Rule for Valid Data Generation
<code>user_id</code>	<code>unsignedBigInteger</code>	Optional; match an existing user ID or leave null
<code>name</code>	<code>string(60)</code>	Optional; full name, ≤ 60 chars or leave blank
<code>email</code>	<code>string(60)</code>	Optional; valid format or blank
<code>subject</code>	<code>string(120)</code>	Short, meaningful topic like "Customer service" (≤ 120 characters)
<code>message</code>	<code>string(250)</code>	Realistic sentence or complaint text (≤ 250 characters)
<code>status</code>	<code>enum</code>	One of: "NEW", "IN_PROGRESS", "RESOLVED"
<code>created_at</code> , <code>updated_at</code>	<code>timestamps</code>	ISO timestamps, <code>updated_at</code> $\geq$ <code>created_at</code>

2.2 Data Generation Tools and Process

To ensure that the generated data was both valid and meaningful, I adopted a semi-automated approach combining AI-assisted data synthesis and structured Python scripting.

2.2.1 Tools Used

1. ChatGPT

Used to synthesize realistic datasets and enrich semantic quality for two key areas:

- **Address-related fields:** (`address`, `city`, `state`, `country`, `postcode`)
- **Contact topics:** Semantic pairs of `subject` and relevant `message` content.

Generated datasets were saved into `.txt` files and then read during the data generation step.

2. Python (Faker library)

Main logic for data generation, combining:

- Field-by-field rule application based on migration and UI constraints.
- Controlled randomness (with seeds) to ensure reproducibility.

Manual `.txt` data sources:

- `subject_message_data.txt` (gened by ChatGPT): Stored 500+ realistic subject-message pairs generated via ChatGPT prompts based on the actual dropdown items seen in the UI.
- `address_data.txt` (gened by ChatGPT): Contained pre-aligned synthetic address entries, also AI-generated, ensuring compatibility between related location fields.

2.2.2 Generation Process

1. Generate Address Data with ChatGPT

Prompt Used for Address Data:

Generate a file .txt contains at least 500 realistic and human-like full address lines for testing purposes.

Each line must include:

- Address
- City
- State
- Country
- Postcode (valid in the selected country)

Requirements:

- All components (address, city, state, country, postal code) must match each other geographically and logically.
- Do not mix locations across countries (e.g., do not put a UK postal code in a US city).
- All lines should be unique and vary across multiple cities and countries.

Format (one line per address):

[address], [city], [state], [country], [postal code]

Example:

123 Maple Street, Springfield, Illinois, USA, 62704

14 King Street, London, England, UK, SW1A 1AA

```
0487 Hull Village Suite 759, New Donald, Florida, Saudi Arabia, 12930
924 Jessica Vista Suite 781, Ryanborough, Oklahoma, Lesotho, 58525
80160 Clayton Isle Apt. 513, East Linda, Georgia, Spain, 71671
15871 Arnold Squares Apt. 858, Port Carrie, Pennsylvania, France, 58826
659 Kelly Field, North Chloe, Texas, Eritrea, 62960
22018 Massey Pine, Aimeebury, Indiana, Ukraine, 89561
477 Miller Ridge Apt. 795, East Allen, Louisiana, Brazil, 92950
2560 Krista Walk, North Lisa, South Dakota, United States of America, 70519
13991 Davis Village, North Catherineborough, Idaho, Saint Barthelemy, 16740
00869 Mary Cliff Apt. 145, Whitehaven, New Jersey, Armenia, 78683
634 Strong Mountains, Lake Emily, Wyoming, Azerbaijan, 89132
58419 Wade Bridge, Maldonadoshire, North Carolina, United States Virgin Islands, 41322
4280 Harper Falls Apt. 842, Port Stephanie, Minnesota, New Caledonia, 28222
```

Figure 1: File address_data.txt

2. Generate Contact Message Data with ChatGPT

Prompt Used for Subject–Message Pairs:

Generate a .txt file contains 500 contact request subject-message pairs for a customer/admin support system.

Each subject must be selected from the following list:

- Customer service
- Webmaster
- Return
- Error 101: Subject not found
- Payments
- Warranty
- Status of my order
- Error 202: Translation error

For each pair:

- Use the subject exactly as written (case-sensitive).
- Write a realistic message (1-3 sentences, not more than 250 characters) that logically relates to the subject.
- Ensure the message sounds like something a real user might write in a contact form.

Format:

[subject]: [message]

Example:

Payments: I made a payment two days ago but it's not showing in my order history.

```
Webmaster: Few almost example collection worry mean prove. International chance despite series alone close.
Payments: Force record star him high firm. International at north hand because great situation station.
Error 101: Subject not found: Half time join purpose oil shoulder become. Include ready election sea. Meet charge human nor
Webmaster: Home sit claim unit stock. International point book girl if near.
Status of my order: Power yeah fill.
Customer service: Performance development always pattern reveal increase.
Error 101: Subject not found: Director term traditional grow range compare behavior. Impact middle anyone. Plan human your.
Customer service: Cultural see although effort. Least fund child although husband. System paper turn sort during road sure.
Status of my order: Call in region news point process significant. Red under lawyer. As fall partner trade buy nearly east.
Error 202: Translation error: Type reduce act home stuff no deep. Six later safe natural bit win. Go measure process.
Customer service: Occur hold wonder else. Entire image assume example law.
Status of my order: Wear Congress by it. Sell election Mr left though really. How concern red executive item develop wind. Y
Payments: Until at turn example wear officer. Put still nothing event.
Error 101: Subject not found: Table popular language partner can space still. Technology today there page for. Traditional s
Webmaster: Surface kitchen majority go friend would player. So a apply short effect finally what may.
Status of my order: Successful point stand people reveal. Relate with issue actually.
```

Figure 2: File subject_message_data.txt

3. Implement Custom Generator in Jupyter Notebook

A custom .ipynb script generates 500 rows for both users and contact_requests tables. The process followed these key steps:

- **Step 1:** Read address_data.txt and split each line into components: address, city, state, country, and postcode.
- **Step 2:** Read subject_message_data.txt and split into paired fields: subject and corresponding message.
- **Step 3:** For each record (iteration 1 to 500):
 - Randomly choose an address line and decompose into fields.
 - Generate realistic user data using **faker** (first name, last name, phone, email, date of birth).
 - Randomly assign optional fields as blank/null based on nullability.
 - Ensure logical constraints, such as:
 - * updated_at greater than created_at

- * `dob` between 1960 and 2010
- * unique emails
- **Step 4:** For contact requests:
 - Randomly choose an existing user ID.
 - Randomly assign subject–message pairs from the dataset.
 - Randomly assign one of the valid `status` enum values: `"NEW"`, `"IN_PROGRESS"`, `"RESOLVED"`

All timestamps are generated using `faker.date_time_this_decade()` to ensure recent but valid ISO timestamp formats.

4. Output and Export

At the end of the process:

- All generated data was structured into two `pandas.DataFrames`.
- The entire dataset was exported into a single Excel file contains two sheets: `users` and `contact_requests`.

2.2.3 Sample Data

1. Table Users

id	first_name	last_name	address	city	state	country	postcode	phone	dob	email	password	role	enabled	_login_atte	created_at	updated_at
1	Billy	Sheppard	924 Jessica Ryanborou	Oklahoma	Lesotho		58525	647593824	1985-09-06	kyleblair@i	B23SKQ%	admin	TRUE		3 2023-07-30T16:05:25	2023-08-26T16:05:25
2	John	Nguyen	80160 Clay East Linda	Georgia	Spain		71671	393328711	1993-01-21	andrea18@	3tlB51m#	admin	TRUE		4 2022-11-01T00:21:22	2022-11-02T00:21:22
3	Frances	Massey	15871 Arn Port Carrie	Pennsylvan	France		58826	833969477	2007-01-21	tuckercryst	S0kim*	d_s admin	TRUE		2 2024-08-06T14:03:07	2024-08-21T14:03:07
4	David	Davis	659 Kelly F North Chlo	Texas	Eritrea		62960	09032173C	2000-09-24	huffchad@	_PL*znEgot	admin	TRUE		2 2024-11-24T10:09:53	2024-12-24T10:09:53
5	David	Morse	22018 Mas Aimeebury	Indiana	Ukraine		69561	076984564	1995-07-03	rogersbrani	E93YpeGlc	customer	TRUE		4 2022-02-17T20:36:10	2022-03-07T20:36:10
6	Jennifer	Nunez	477 Miller East Allen	Louisiana	Brazil		92950	069602714	1989-12-15	hensontroy	l3xyn@w*	admin	TRUE		1 2023-03-22T10:02:47	2023-03-26T10:02:47
7	Charlene	Gonzalez	2560 Krista North Usa	South Dakc	United Stat		70519	421761047	1972-06-06	darin24@e	780GoovW	customer	TRUE		4 2021-01-04T10:07:01	2021-01-28T10:07:01
8	Alan	Hernandez	13991 Davi North Cath	Idaho	Saint Barth		16740	551590042	1999-06-05	ihays@exaiv	pfAXAh	admin	TRUE		4 2022-06-23T05:10:45	2022-07-18T05:10:45
9	Benjamin	Rogers	00869 Mar Whitehave	New Jersey	Armenia		78683	29611133C	1993-05-05	shane79@i	F_7j3ENcd	customer	TRUE		2 2023-08-18T04:46:44	2023-09-09T04:46:44
10	Bobby	Gould	634 Strong Lake Emily	Wyoming	Azerbaijan		69132	921376582	1973-07-28	blittle@exi	Yq5uPepB	customer	TRUE		5 2025-06-14T12:13:15	2025-06-17T12:13:15
11	Carl	Miller	58419 Wac Maldonadc	North Caro	United Stat		41322	132040752	2008-05-04	gsantana@	jizt3RWn1	admin	TRUE		4 2024-01-01T13:54:22	2024-01-11T13:54:22
12	Kevin	Hudson	4280 Harm Port Stephi	Minnesota	New Caled		38233	174466602	1959-06-27	mcmahonti	b9ziXGSM*	admin	TRUE		3 2021-07-19T08:35:10	2021-07-22T08:35:10
13	Patrick	Frank	924 Goodm Ernestbush	South Dakc	Kiribati		50131	109795194	1983-01-23	benjaminm	Yws#6@zU	customer	TRUE		4 2021-08-28T18:17:12	2021-09-07T18:17:12
14	Sonia	Thomas	69602 Brov North Troy	New Jersey	Australia		54913	116116207	1993-12-24	jennifer15@	1n5Wa_	admin	TRUE		4 2022-08-06T13:40:55	2022-08-21T13:40:55
15	Michele	Lopez	120 Herring Aarontown	Virginia	Saudi Arab		00810	754203557	2000-08-18	brittannym	sl48FhmJ5	customer	TRUE		4 2021-12-24T11:31:53	2022-01-01T11:31:53
16	Joyce	Thomas	19344 Crai Burtontow	Louisiana	United Stat		59873	45137580E	1996-12-21	sharris@ex	8@cZLHid	customer	TRUE		0 2023-07-18T19:04:54	2023-08-16T19:04:54
17	Maria	Cole	2851 Joshu North Jami	Oklahoma	Germany		48590	03414892E	1979-07-11	bramirez@	H0eAhiqb%	admin	TRUE		5 2020-01-06T17:26:09	2020-01-29T17:26:09
18	Robert	Roberts	977 Watso East Nichol	Maine	Algeria		18648	983816864	1970-08-06	ericterry@	i&AYO4eP	O customer	TRUE		4 2020-07-30T13:22:46	2020-08-25T13:22:46
19	Pamela	Smith	5551 Chan West Frank	Maine	Guyana		52247	160501904	1991-07-23	james67@	JsK*eQfKl	admin	TRUE		1 2023-04-29T16:32:28	2023-05-14T16:32:28
20	Jennifer	Bullock	1730 Whit West Richa	Maryland	Israel		14742	312502394	1973-05-16	robert08@	UOeX75n_	admin	TRUE		5 2024-12-27T06:51:56	2025-01-19T06:51:56
21	Matthew	Garcia	775 Martin Simmons	Michigan	Saint Pierre		90561	55875408C	1990-05-06	jacob34@e	p+4JSv*qs	customer	TRUE		1 2021-08-22T03:48:24	2021-09-18T03:48:24
22	Scott	Jones	96111 Kara Smithfur	Oklahoma	Malaysia		38488	860237324	1987-08-24	chelsea34@	9@Xe0r4r	customer	TRUE		1 2021-10-25T05:13:04	2021-11-23T05:13:04
23	Teresa	Cunningham	9361 Robin North Lynn	North Dakc	Korea		99124	643318453	1978-01-05	derricksmi	@5P^uujl	customer	TRUE		4 2021-01-02T08:56:04	2021-01-27T08:56:04
24	Brittany	Fuller	76430 Cind South Nich	Ohio	Sweden		20336	90895774C	1983-07-11	tpayne@ex	em2GTmf	customer	TRUE		0 2025-03-05T12:58:48	2025-03-19T12:58:48
25	Michael	Ryan	729 Little A Sheltonm	New York	Luxembour		83740	507003003	1980-03-21	ntaylor@e	&ZlHgljR	admin	TRUE		0 2020-02-22T17:51:40	2020-03-03T17:51:40
26	Christy	Stanley	930 Thornt Lisaton	North Dakc	Cameroon		34171	153230282	1993-07-14	nmckinney	lFYr4EkzP	admin	TRUE		5 2021-03-05T10:58:04	2021-03-14T10:58:04
27	Andrea	Madden	9471 Zachs East David	Alabama	Senegal		59298	07616570C	1995-07-24	wanda38@	ylv5gRq	admin	TRUE		4 2022-06-14T11:48:58	2022-06-17T11:48:58
28	Connie	Morris	27586 Aud Petersenfo	Missouri	Netherlands		77360	962029072	1984-03-05	chadboyle@	a+18Fch7#J	admin	TRUE		3 2022-06-23T16:44:45	2022-06-29T16:44:45

Figure 3: Users Sheet

2. Table Contact Requests

id	user_id	name	email	subject	message	status	created_at	updated_at
1	398	William Woodard	ecarney@examp	Payments	Force record star him high firm. Ir	RESOLVED	2025-04-29T23:55:22	2025-04-30T00:51:22
2	345	Kristen Peterson	vsmith@exampl	Error 101	Subject not found: Half time join	IN_PROGRESS	2025-01-23T02:42:53	2025-01-23T04:03:53
3	451	Cameron Rocha	nroman@exam	Webmaster	Home sit claim unit stock. Interna	NEW	2025-05-26T21:12:30	2025-05-26T22:42:30
4	239	Tina Howell	thampton@exa	nStatus of m	Power yeah fill.	IN_PROGRESS	2025-02-21T20:39:43	2025-02-21T21:54:43
5	292	Chelsea Forbes	jasonjoseph@ex	iCustomer s	Performance development always	NEW	2025-02-21T22:39:27	2025-02-22T00:26:27
6	364	Brian Chung	warderin@exam	Error 101	Subject not found: Director term	IN_PROGRESS	2025-01-21T02:56:36	2025-01-21T04:09:36
7	421	Lisa Davis	christinaavery@	eCustomer s	Cultural see although effort. Least	NEW	2025-02-09T06:52:23	2025-02-09T07:34:23
8	243	Dennis Soto	martindiana@ex	Status of m	Call in region news point process	RESOLVED	2025-05-16T05:41:25	2025-05-16T06:32:25
9	22	Adam Fernandez	angela67@exam	Error 202	Translation error: Type reduce act	NEW	2025-01-13T13:44:59	2025-01-13T14:02:59
10	443	Jimmy Smith	donald59@exam	Customer s	Occur hold wonder else. Entire im	IN_PROGRESS	2025-02-20T13:04:08	2025-02-20T14:17:08
11	3	Ashley Nelson	trevor07@exam	Status of m	Wear Congress by it. Sell election	NEW	2025-06-09T07:47:31	2025-06-09T08:34:31
12	368	Michael Coleman	brandonward@e	Payments	Until at turn example wear office	NEW	2025-02-09T19:45:36	2025-02-09T20:10:36
13	277	Molly Ruiz	joseph91@exam	Error 101	Subject not found: Table popular	IN_PROGRESS	2025-05-01T22:54:41	2025-05-02T00:20:41
14	162	Victoria Lowe	librenda@exam	Webmaster	Surface kitchen majority go frienc	NEW	2025-05-09T06:06:33	2025-05-09T06:12:33
15	202	Walter Reed	joelthomas@exa	Status of m	Successful point stand people rev	NEW	2025-06-04T13:28:20	2025-06-04T14:09:20
16	326	Brenda Benton	emccullough@ex	Warranty	State official but work none black	IN_PROGRESS	2025-03-13T04:13:32	2025-03-13T05:51:32
17	344	Jackson Aguirre	danielrojas@exa	Payments	Positive east thousand director sc	NEW	2025-05-18T12:52:35	2025-05-18T13:45:35
18	76	Brandon Ford	wayneromero@e	Error 202	Translation error: Group simply m	IN_PROGRESS	2025-06-14T23:49:55	2025-06-15T01:06:55
19	157	Jesse Marshall	shelbymoore@e	Status of m	Traditional hit rule in. Away push	RESOLVED	2025-05-03T16:45:45	2025-05-03T18:36:45
20	84	Thomas Hughes	boylemartin@ex	Payments	Product stuff public exist stock cle	NEW	2025-04-18T22:00:31	2025-04-18T22:08:31
21	69	Jamie Ingram	qmaynard@exan	Error 101	Subject not found: Test when lool	IN_PROGRESS	2025-04-14T18:53:08	2025-04-14T20:48:08
22	329	Jordan Harris	jburnett@examp	Customer s	Here take foot film same. Thousai	RESOLVED	2025-02-08T09:50:53	2025-02-08T11:21:53
23	371	Abigail Bell	xyoung@exampl	Payments	Piece site fund international door	NEW	2025-03-26T09:17:48	2025-03-26T10:55:48
24	422	Jacqueline Johnson	michael45@exar	Error 101	Subject not found: South cost live	RESOLVED	2025-02-23T10:46:15	2025-02-23T12:20:15
25	422	Jesse George	branchjessica@e	Status of m	Push later paper official. In educa	RESOLVED	2025-01-16T22:48:35	2025-01-16T22:54:35
26	382	Robert Horton	jessica85@exam	Error 202	Translation error: Another variou	RESOLVED	2025-01-24T10:45:31	2025-01-24T12:14:31
27	199	Tammy Cox	katherinewhite@	Return	Score daughter seem ok. Radio hc	NEW	2025-05-08T22:49:43	2025-05-09T00:10:43
28	410	Jennifer Daniels	nrivas@example	Warranty	Owner foreign ground campaign i	RESOLVED	2025-05-09T09:43:22	2025-05-09T10:28:22

Figure 4: Contact Requests Sheet

3 Self-Evaluation (Data Generation)

Criteria	Self-Evaluation	Notes
2 Tables Selection	1.0 / 1.0	Selected two important tables: <code>users</code> and <code>contact_requests</code> .
Sample Data	2.0 / 2.0	All data is meaningful, realistic, and aligned with UI/business context.
Data Generation Report	1.0 / 1.0	Report includes field rules, tools used, prompts, process, and samples.